



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

FLASH COLUMN BOTTOMS COOLER (E-2306)

Docu. No. : 14780-8120-PS-E2306
Project No. : 120117
Rev. No. : Z1
Date : MAY 11, 2016
Page : 1 / 10

TITLE :

FLASH COLUMN BOTTOMS COOLER

(E-2306)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-E2306
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 11, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 8, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 11, 2016	MAY 11, 2016	-	MAY 11, 2016	

This Document is the property of POSCO Engineering Co., Ltd. Therefore, it shall not be released to Any third party without permission of an authorized personnel of the POSCO Engineering Co., Ltd.

[illegible]



SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units **Metric**

Doc. No. : 14780-8120-PS-E2306

Rev. No. : Z1

Item No. : E-2306

Page : 3 / 10

								Rev
Case	Design	TEMA Type (Note 2)	C_U	Orientation	Vertical			
Number of Units	1	Shells per Unit	1	Shells in Series	1			
Surface per Unit	662.8	Surface per Shell	662.8	Shells in Parallel	1			
		(Note 12)		(Note 12)				
PERFORMANCE OF ONE UNIT								
		In	SHELLSIDE	Out	In	TUBESIDE	Out	
Fluid Name		Cumene & CHP			Reliable Cooling Water			
Total Flowrate (Note 3)		kg/h	92836 × 1.05			150578 × 1.05 (Note 20)		
Vapour		kg/h	0	0		0	0	
Liquid		kg/h	92836 × 1.05	92836 × 1.05		150578 × 1.05	150578 × 1.05	
Fluid Condensed		kg/h	0			0		
Fluid Evaporated		kg/h	0			0		
LIQUID PROPERTIES								
		SEE PAGE 8			AS WATER			
Density	kg/m³							
Viscosity	cP							
Specific Heat	kcal/kg.°C							
Thermal Conductivity	kcal/h.m.°C							
Surface Tension	dyne/cm							
Pour Point Temperature	°C							
VAPOUR PROPERTIES								
Density	kg/m³							
Viscosity	cP							
Specific Heat	kcal/kg.°C							
Thermal Conductivity	kcal/h.m.°C							
Hydrate Temperature	°C							
THERMAL AND HYDRAULIC DATA								
Operating Temperature	°C	83.3	60		33.0	38		
Operating Pressure	kg/cm²g	-0.2375			3.9			
Number of Passes per Shell		1			16			
Average Velocity	m/s	0.06			1.1			
Pressure Drop (Allow / Calc)	kg/cm²	Nil	0.006		0.703 (Note 21)	0.722		
Fouling Factor	h.m².°C/kcal	0.0003			0.0004			
Duty (Note 3)	kcal/h × 10⁶	0.7518 × 1.05			MTD	°C	34.1	
U-value Service/Clean	kcal/h.m².°C	83	89		Over Surface	%	139	
								(Notes 11, 12)
DESIGN CONDITIONS								
		SHELLSIDE			TUBESIDE			
Design pressure max/min	kg/cm²g	3.5			10.5			
Full Vacuum		Required	at 195 °C		Not Required			
Test Pressure	kg/cm²g	To Code			To Code			
Design Temperature max/min	°C	250	15		120	15		
Corrosion Allowance	mm	1.5			3.2	(For KCS)		
Insulation Details		TBA			None			
Hydrogen Service		No			No			
Sour Service		No			No			
Lethal Service		No			No			
Cyclic Service		No			No			
Cleaning Method		Mechanical			Mechanical			
Stress Relief for Process Reasons		No			No			
Heat Release Curve		Yes			No			



SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

Doc. No. : 14780-8120-PS-E2306

Rev. No. : Z1

Item No. : E-2306

Page : 4 / 10

CONSTRUCTION OF ONE SHELL

SHELL	Shell I.D.	mm	2200 (Note 18)	Channel I.D. (If different)	mm	2200
TUBES	Number of Tubes		3606 (Note 4)	Tube Length	mm	2438
	Tube O.D.	mm	19.05	Tube Thickness	mm	1.651
	Tube Gauge	BWG	16	Tube Wall Basis		Average
	Tube Pitch	mm	30.8	Tube Layout Angle	deg.	90
	Tube-Tubesheet Joint		Strength Welded (Note 9)	Tube-Baffle Diam. Clearance	mm	0.8
BAFFLES	Number of Baffles		4	Baffle Type		Single Seg.
	Baffle Cut	%	18.2	Baffle Spacing (Note 7)	mm	440 (c/c)
	Baffle Cut Orientation		Horizontal	Double Seg. Baffle Overlap	mm	0
	Baffle Thick	mm	12.7	Baffle-Shell Diam. Clearance	mm	11.1
BUNDLE	Impingement Protection		None	Tube Supports		None
	Pairs of Sealing Strips		0	Sealing Strip Details		None
	Passlane Sealing		Yes	Passlane Sealing Details (Note 6)		Maximum Length
	U-Bend Diam.	mm	57.15	U-Bend Orientation		Horizontal
	U-Bend Area Included		No	U-Bend Support Details		(Note 5)
	Longitudinal baffle		No	Longitudinal Baffle Seal		N.A.

CONNECTIONS

SHELLSIDE						TUBESIDE					
Service	No.	N.B.		Rating	Facing	Service	No.	N.B.		Rating	Facing
Inlet S1	1					Inlet T1	1	6	Inch	150	RFWN
Outlet S2	1	10	Inch	150	RFWN	Outlet T2	1	6	Inch	150	RFWN
Drain SD	1	3/4	Inch	150	RFLWN	Vent TV	1	3/4	Inch	150	RFLWN
						Drain TD	1	2	Inch	150	RFLWN

MATERIALS

Tubes	SA 213 TP304L
Shell / Shell Cover	N.A.
Floating Head Cover	N.A.
Channel / Channel Cover	SA 240 TP304L / SA 182 F304L (UOP Std 3-15) (Note 8) or SA 516 Gr. 60N / SA 266N Gr. 2 (Killed) (UOP Std 3-11) (Note 8)
Tubesheet, Stationary	SA 182 F304L
Tubesheet, Floating	N.A.
Baffles / Tube Supports	A 240 TP304
Nozzles / Nozzle Flange	Shellside: N/A
Nozzles / Nozzle Flange	Tubeside: SA 312 TP304L / SA 182 F304L
Bolt and nuts	SA 193 B8 / SA 194-8
Gaskets, Shellside	Grooved Metal Gasket 316L SS with Graphite covers
Gaskets, Tubeside	Grooved Metal Gasket 316L SS with Graphite covers
Gaskets, Nozzles	Spiral Wound SS 316, Non Asbestos, CS centering / SS 316 inner ring

ESTIMATED WEIGHTS

Weights per Shell	Exchanger Weight, Dry	TBA	Exchanger Weight, Wet	TBA
(Tonnes)	Bundle Weight, Dry	11.3		

DESIGN REQUIREMENTS

PED Classification	Shellside- N.A.	Tubeside- N.A.
Project Specifications	UOP 963766 - 401, UOP Std 4-11-7, UOP Std 3-11, UOP Std 3-15, 14780-21A1	
Design Code	ASME VIII Div 1	
Equipment Code	TEMA Class R 9th edition, API 660 8th edition 2007	



SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

Doc. No. : 14780-8120-PS-E2306

Rev. No. : Z1

Item No. : E-2306

Page : 5 / 10

Notes:

1. Item has been checked for Flow Induced Vibration. Tube damage is not predicted.
2. Exchanger will be installed vertically into the bottom of Flash Column, V-2302 (Refer to UOP 963766-301). See page 7.
Based on detail "A" on page 9, CONTRACTOR to revise the drawing for a TEMA type C front head. OWNER/CONSULTANT approval is required if a TEMA type A front head is used.
3. Flow and duty include 5% design margin.
4. For U-tube design, Number of tubes equals number of straights.
5. Provide one In bend U-bend support to hold 33 outer most tube rows.
6. Seal rods and bundle runners shall not restrict the bundle cleaning lanes.
7. The outlet spacing is 1088.5 mm.
8. For the TEMA type C front head, the channel material shall be the same as the tubesheet.
For the TEMA type A front head if selected, the channel material shall be Killed Carbon Steel similar to the channel cover material. UOP Std 3-11 shall be applied.
9. Strength-weld tubes to tubesheet per requirements as specified on page 10.
10. Maximum diametral clearance between bundle outer tube limit diameter and shell (Column) inside diameter shall be 25.4 mm. CONTRACTOR/VENDOR to confirm.
11. Based on LMTD of 24.4°C for equal hot side temperature of 60°C at inlet and outlet,
% Over surface is 71%.
12. Based on bottom 2 m of tube length to handle total duty, the effective area is 412.2 m² and
% Over surface is 0.3% at LMTD of 24.4°C. CONTRACTOR / VENDOR to confirm.
13. The maximum allowable leakage of cooling water into the process is 100 wt ppm.
VENDOR to confirm.
14. CONTRACTOR shall confirm the mating flange sizes. Provision for complete vessel (shell side) draining is required. (2" Nozzle)
15. Fire water may be used as an emergency coolant. The tube side design conditions must be confirmed by CONTRACTOR.
16. Additional requirements and design notes on page 9 shall be applied.
17. Bundle runners shall not restrict the shell side exit flow path.
18. CONTRACTOR to provide vessel flange information/drawing by Exchanger VENDOR to Vessel VENDOR for flanges alignment.
19. A Magnesium anode shall be provided on the tubeside.
20. To maintain the minimum tubeside velocity, CONTRACTOR to confirm that the cooling water flowrate is available.
21. The allowable tubeside pressure drop to be confirmed by CONTRACTOR.
22. Deleted.

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

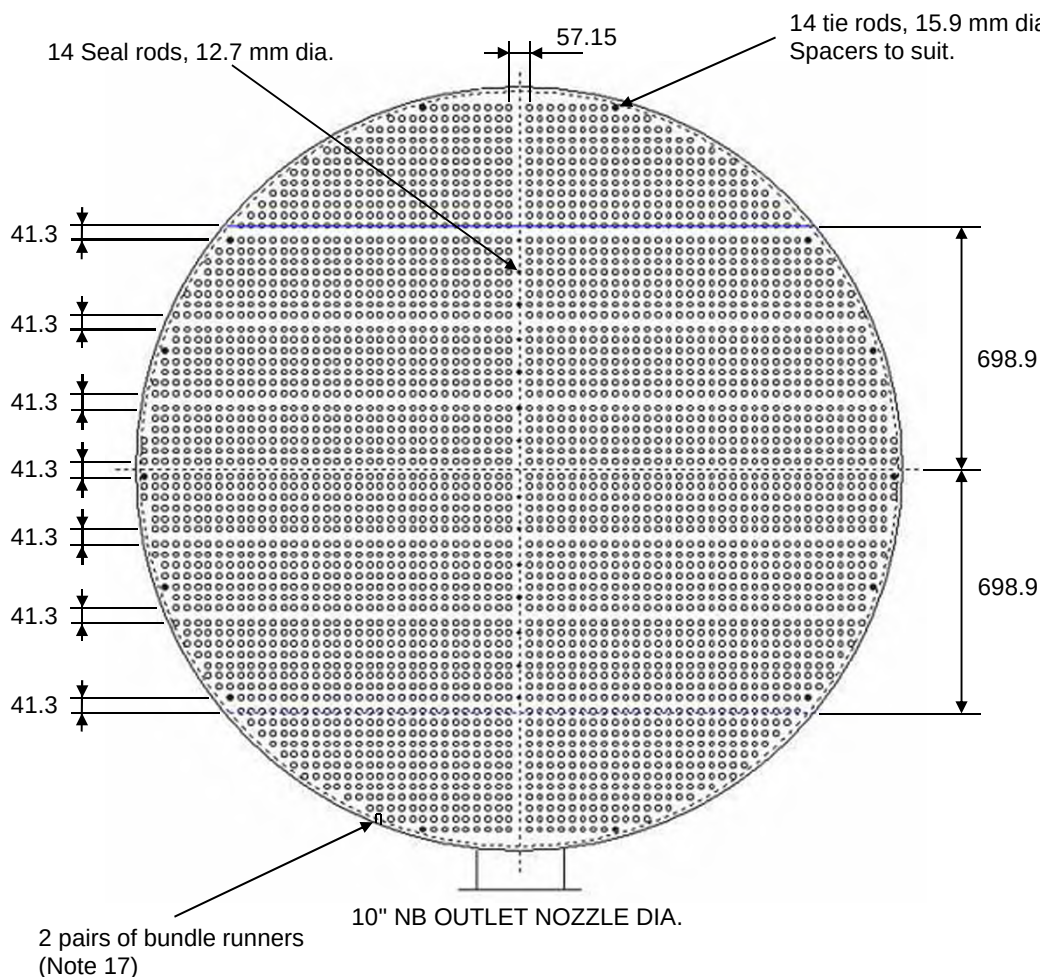
Doc. No. : 14780-8120-PS-E2306

Rev. No. : Z1

Item No. : E-2306

Page : 6 / 10

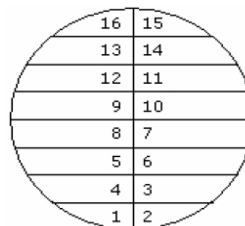
Tube Layout



O.T.L. = 2176.9 mm (Note 10)

All dimension in mm

Sketch not to scale



Tubepass Arrangement

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

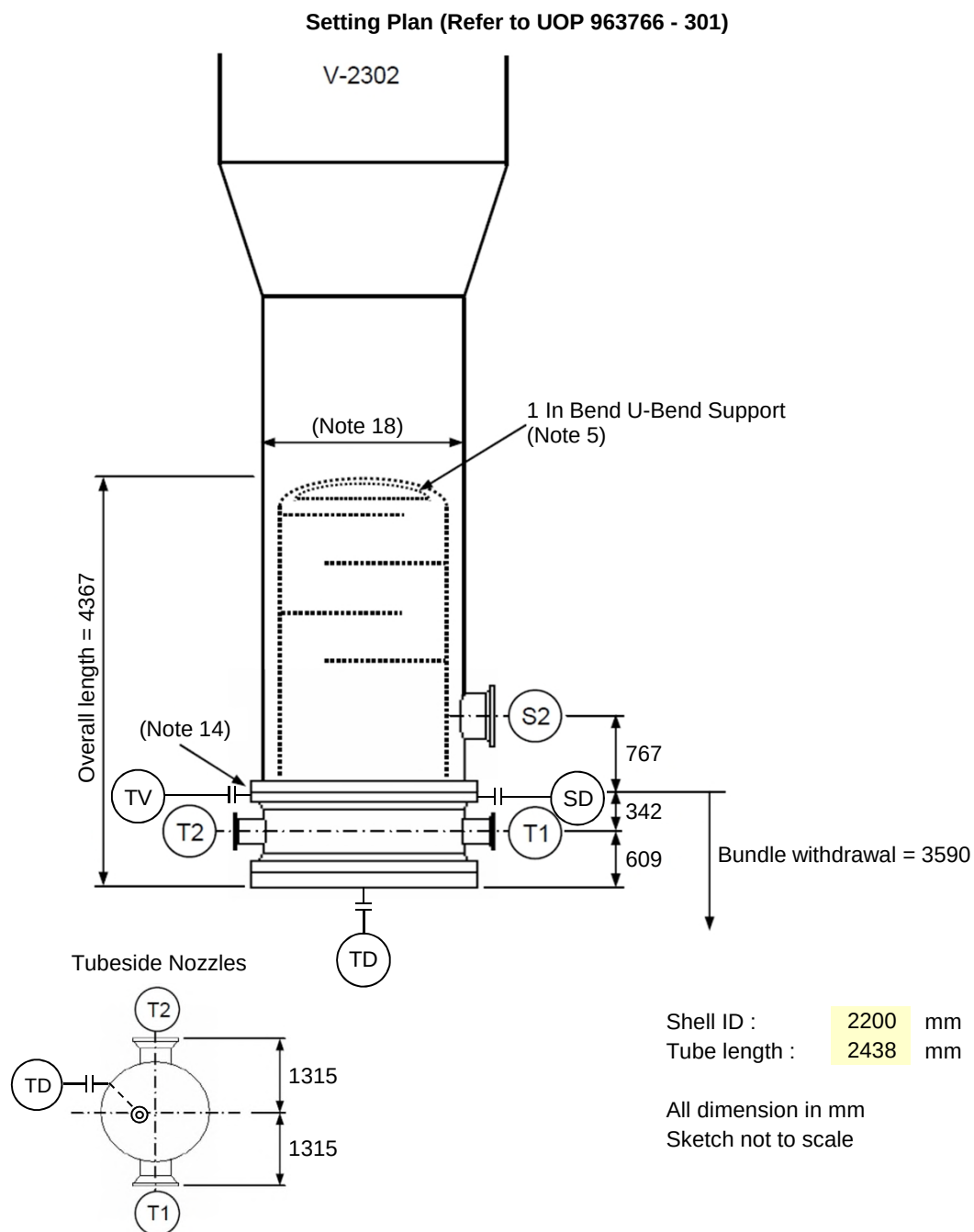
System of Units **Metric**

Doc. No. : 14780-8120-PS-E2306

Rev. No. : Z1

Item No. : E-2306

Page : 7 / 10



PROCESS NOZZLE SUMMARY					
REF	SERVICE	SIZE	RATING	ELEV.	REMARK
S1	Cumene & CHP Inlet				
S2	Cumene & CHP Outlet	10"	150#RFWN		
T1	Reliable Cooling Water Inlet	6"	150#RFWN		
T2	Reliable Cooling Water Outlet	6"	150#RFWN		
SD	Shellside Drain	3/4"	150#RFLWN		CW/blind
TV	Tubeside Vent	3/4"	150#RFLWN		CW/blind
TD	Tubeside Drain	2"	150#RFLWN		CW/blind

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

Doc. No. : 14780-8120-PS-E2306

Rev. No. : Z1

Item No. : E-2306

Page : 8 / 10

HotSide Property Curve

Pressure

-0.2375

kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
83.3	8.09	0.0000	-	-	-	-
80.8	7.19	0.0000	-	-	-	-
78.2	6.29	0.0000	-	-	-	-
75.7	5.39	0.0000	-	-	-	-
73.1	4.50	0.0000	-	-	-	-
70.5	3.60	0.0000	-	-	-	-
67.9	2.70	0.0000	-	-	-	-
65.3	1.80	0.0000	-	-	-	-
62.7	0.90	0.0000	-	-	-	-
60.0	0.00	0.0000	-	-	-	-

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
83.3	946.8	2.050	0.073	0.36	20.7	33.9	338
80.8	949.4	2.163	0.073	0.36	20.9	33.9	338
78.2	952.1	2.285	0.074	0.35	21.1	33.9	338
75.7	954.8	2.416	0.074	0.35	21.4	33.9	338
73.1	957.5	2.558	0.075	0.35	21.6	33.9	338
70.5	960.2	2.712	0.075	0.35	21.9	33.9	338
67.9	962.9	2.878	0.075	0.34	22.1	33.9	338
65.3	965.6	3.059	0.076	0.34	22.4	33.9	338
62.7	968.3	3.256	0.076	0.34	22.6	33.9	338
60.0	971.1	3.470	0.077	0.34	22.9	33.9	338

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

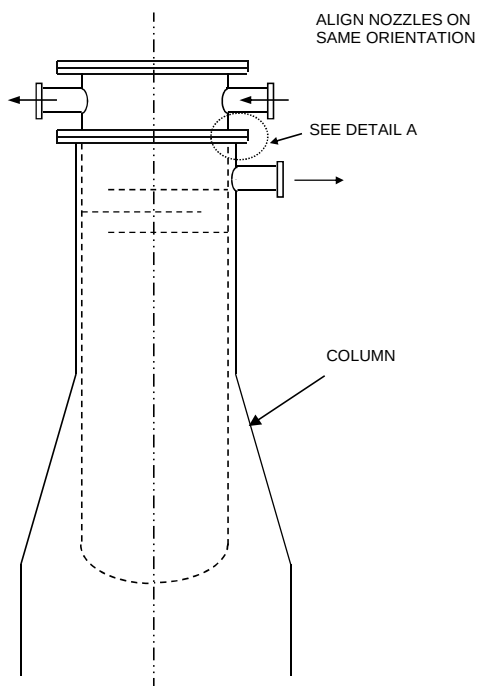
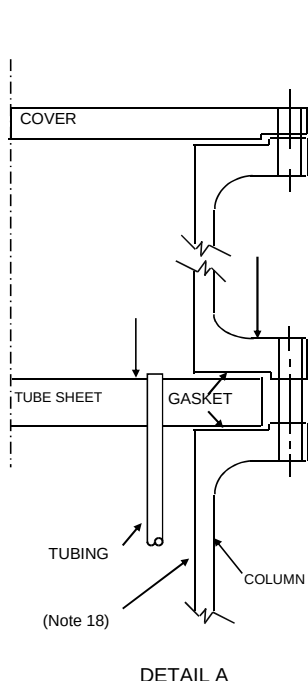
Doc. No. : 14780-8120-PS-E2306

Rev. No. : Z1

Item No. : E-2306

Page : 9 / 10

VERTICAL STABBED-IN CONDENSER/COOLER REQUIREMENTS (Refer to UOP 963766 - 401)



1. REQUIREMENTS

- Bundle vendor shall provide the necessary welding neck flange to column fabricator for attachment to column nozzle.
- Responsibility for design of detail "A" closure is fully chargeable to tube bundle designer except it shall be the TEMA "R" confined gasket type.
- Bundle vendor shall include stationary head together with necessary bolts, nuts and gaskets for securing bundle to column nozzle.

2. TUBE BUNDLE DESIGN:

- The general design shall conform to the operating conditions given on this item's specification sheet.
- In case vendor finds it necessary to increase the nozzle size for guaranteeing the desired process outlet temperature it shall be their responsibility to disclose, prior to start of column design work, the pertinent information that permits designing that portion of column the bundle fits in that assures clearances equal to TEMA "R" at baffle circumference.
- When a TEMA "C" type front head is specified, the bolting of the channel and integral tubesheet to the column nozzle flange shall be typical of a TEMA "C" type head to a shell.

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

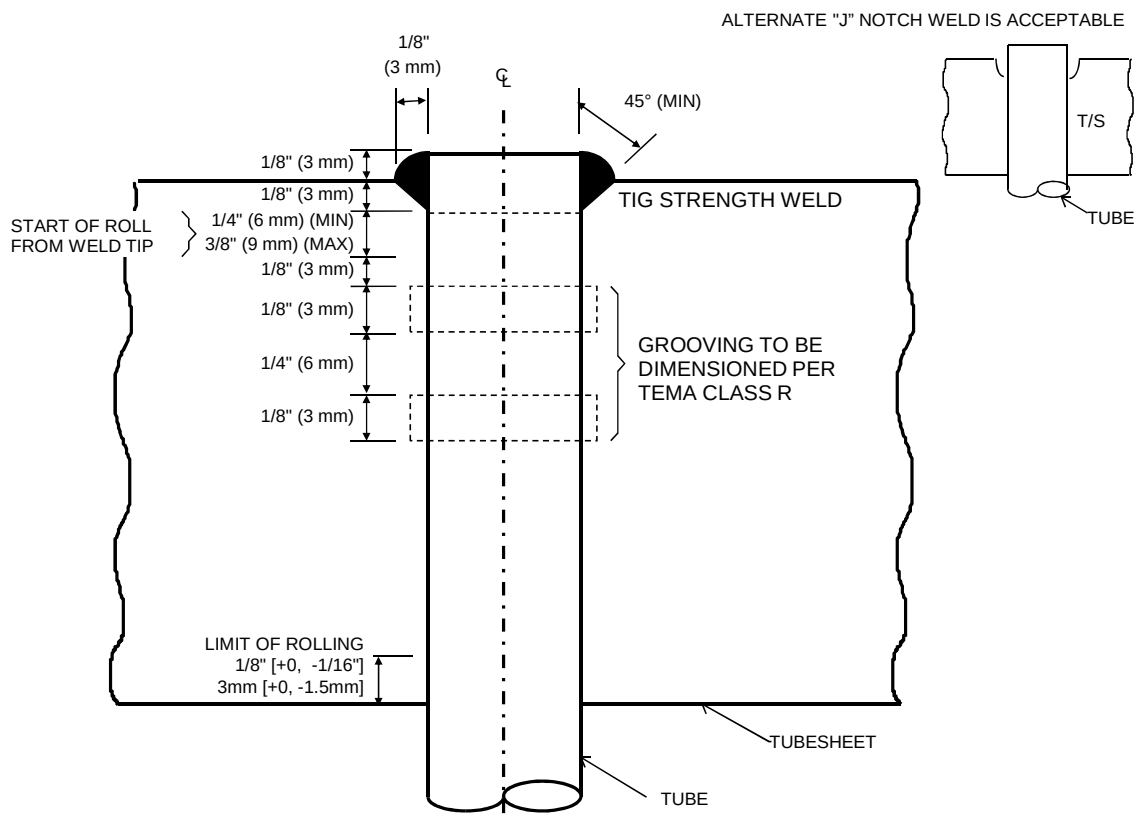
Doc. No. : 14780-8120-PS-E2306

Rev. No. : Z1

Item No. : E-2306

Page : 10 / 10

Strength-Weld Type Tube to Tubesheet Joint Requirements (Refer to UOP 963766 - 401)



1. Tubing shall be seamless type.
2. Joint components shall be deburred and meticulously cleaned prior to welding.
3. When heat treatment of the tubesheet is required, final tube hole sizing shall be attained by reaming after heat treatment.
4. Preheat tubesheet to a minimum of 150°F (65°C) and maintain this temperature during welding.
5. Each tube end shall be welded individually without stopping.
6. Tube to tubesheet joints shall be welded with a minimum of two passes. The first pass shall be 100% examined by dye penetrant or magnetic particle method. Remaining passes shall be welded without interruption. The final welds shall be 100% examined by dye penetrant or magnetic particle method.
7. All tube joint welds shall be proven using a soap test [with minimum 15 psi (1.05 kg/cm²) air] prior to rolling.
8. A helium leak test shall be performed in accordance with the requirements of ASME code, Section V, Article 10. The shell side shall be pressurized with helium while the exchanger is inclined to provide good flow of helium to a single collection point on the tube side. Any leakage in excess of 1×10^{-4} standard cc/sec warrants repair.
9. Do all tube rolling after completion of welding and leak testing.
10. The tubes shall be rolled for the entire tubesheet thickness. Stop shall be used on tube roller that assures expansion is within limits given on sketch.
11. A certified test report of the final, successful helium test shall be provided and signed by an independent third party witness.
12. All applicable Weld Procedure Specifications (WPS) and Welder Qualification Records (WQR) shall be provided per clients request.

Rev